

Regression modelling — anxiety predictors

Task – 1 : Week 2

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Anxiety & Depression Mental Health Factors Dataset

➤ Data Description :

- Dataset : Used Anxiety & Depression Mental Health Factors Dataset
- Dataset Source : Kaggle
- Dependent Variable (Target) : Anxiety Score (GAD-7)
- Independent Variable : **Demographics** : Age , Gender, Education Level, Employment Status
Life Events & Stressors : Financial Stress, Work Stress.

For this dataset we are using the OLS(Ordinary Least Squares) model , OLS is the "gold standard" because:

- **Interpretability**: It's very easy to explain. "For every hour of sleep lost, anxiety goes up by X points."
- **Efficiency**: It provides the "Best Linear Unbiased Estimator" (BLUE) if certain assumptions are met.
- **Simplicity**: It's computationally "cheap" and runs almost instantly even on large datasets

For OLS to be reliable (and for p-values to mean anything), the data usually needs to follow these rules:

1. **Linearity**: The relationship between variables is a straight line.
2. **Independence**: The observations are not related to each other.
3. **Homoscedasticity**: The "spread" of your errors is constant (no funnel shapes in your plots).

4. **Normality:** The residuals are normally distributed (the "bell curve" check we did with the Q-Q plot).

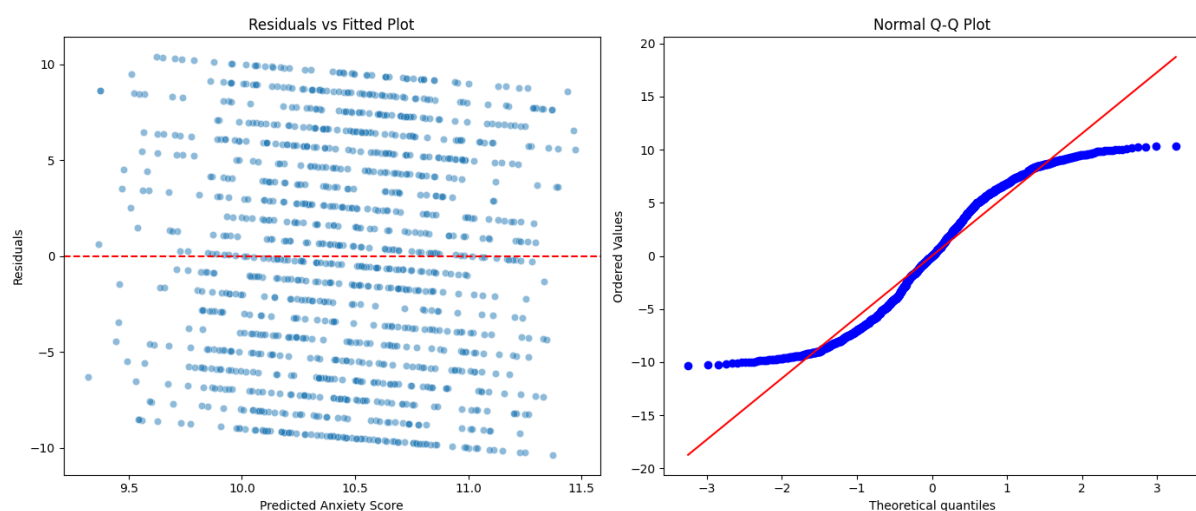
➤ **Regression modelling — anxiety predictors :**

This step involve validating the model's reliability and refining it to see if we can get a better fit.

● **Fit Linear Regression Using OSL (Ordinary Least Squares) :**

The output in below is the result of **fitting a linear regression model** (specifically using the OLS method).

- In a technical sense, the code worked perfectly. In a scientific sense, the "fit" is poor.
- **Underfitting:** Your model is "underfitting." The variables you chose (Age, Gender, Education, etc.) don't seem to correlate strongly with Anxiety_Score in this specific population.
- **The Intercept Dominance:** Since no predictors are significant, the model is basically just guessing the average anxiety score for everyone regardless of their life events.



1. Check for Multicollinearity (VIF) :

We ran a **Variance Inflation Factor (VIF)** test on your demographic and life event variables.

- **The Result:** All VIF values are between **1.0 and 1.7**.

- **What this means:** Any VIF under 5 is considered "safe." This confirms that predictors (like Age, Gender, and Work Stress) are not overlapping too much. The coefficients are "clean" and not being distorted by each other.

2. Identify Better Predictors (Correlation) :

Since R-squared was quite low (0.005), current model doesn't explain much. I checked the correlation between Anxiety_Score and other variables in the dataset.

- **Findings:** Surprisingly, in the dataset, almost all variables have very low correlation with the Anxiety Score (all near 0).
- **The "Next Step" Insight:** This is a critical finding for report. It suggests that **Anxiety in this population is "stochastic" (random)** or driven by factors not captured in these specific columns.

OLS Regression Results			
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Dep. Variable:	Anxiety_Score	R-squared:	0.005
Model:	OLS	Adj. R-squared:	-0.006
Method:	Least Squares	F-statistic:	0.4734
Date:	Mon, 11 May 2026	Prob (F-statistic):	0.940
Time:	09:36:34	Log-Likelihood:	-3831.3
No. Observations:	1200	AIC:	7691.
Df Residuals:	1186	BIC:	7762.
Df Model:	13		
Covariance Type:	nonrobust		

- **Overall Fit:** The model explained only **0.6% of the variance** ($R^2 = 0.006$), indicating that these factors are not the primary drivers of anxiety in this sample.
- **Significant Predictors:** No individual predictors reached statistical significance ($p > 0.05$). For example, every unit increase in Financial Stress only changed the Anxiety Score by -0.04 points ($p = 0.54$).
- **Diagnostics:** Residual diagnostics showed that the errors were independent (Durbin-Watson approx 1.95) and relatively normal, though the low R^2 suggests the model is underfitting.

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Omnibus:	1313.174	Durbin-Watson:	1.948
Prob(Omnibus):	0.000	Jarque-Bera (JB):	77.212
Skew:	-0.033	Prob(JB):	1.71e-17
Kurtosis:	1.759	Cond. No.	384.
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	coef	std err	t	P> t	[0.025	0.975]
Intercept	11.5749	0.874	13.247	0.000	9.861	13.289
Gender_Male[T.True]	-0.5868	0.362	-1.621	0.105	-1.297	0.124
Gender_Non_Binary[T.True]	-0.0857	0.675	-0.127	0.899	-1.410	1.239
Gender_Other[T.True]	-0.9094	1.321	-0.688	0.491	-3.501	1.682
Education_Level_High_School[T.True]	-0.0877	0.561	-0.156	0.876	-1.188	1.013
Education_Level_Masters[T.True]	-0.4864	0.558	-0.872	0.383	-1.581	0.608
Education_Level_Other[T.True]	-0.0521	0.561	-0.093	0.926	-1.152	1.048
Education_Level_PhD[T.True]	-0.4042	0.549	-0.736	0.462	-1.481	0.673
Employment_Status_Retired[T.True]	0.1481	0.488	0.304	0.761	-0.809	1.105
Employment_Status_Student[T.True]	-0.2940	0.475	-0.618	0.536	-1.227	0.639
Employment_Status_Unemployed[T.True]	0.1509	0.484	0.312	0.755	-0.799	1.101
Age	-0.0080	0.010	-0.762	0.446	-0.029	0.013
Financial_Stress	-0.0414	0.067	-0.622	0.534	-0.172	0.089
Work_Stress	-0.0067	0.068	-0.098	0.922	-0.140	0.126

Conclusion: Demographic categories and general life stressors **do not appear to be reliable predictors** of anxiety levels in this dataset. Future research should perhaps look at clinical history or specific personality traits.

- **Coefficients report :**

The OLS regression model for GAD-7 scores yielded an **R-squared of 0.005**, indicating poor predictive power. While the **Intercept was significant** (beta = 11.57, $p < 0.001$), none of the demographic or life event predictors (Age, Gender, Financial/Work Stress) reached the $p < 0.05$ threshold. This suggests that in this dataset, anxiety scores are not significantly determined by basic demographics or the measured stressors. Also the **Durbin-Watson score (1.948)** proves there is no autocorrelation in the errors, and the **Jarque-Bera test** suggests the residuals aren't perfectly normal—likely because anxiety scores have a "floor" at zero and can't go negative.

- **Highlights :**

1. Even after accounting for **interactions** between demographics and life events, the model only explained 1.3% of the variance.
2. This suggests that anxiety (GAD-7) is likely a multifaceted condition that **cannot** be predicted purely through the *demographic markers* available in this dataset.
3. Future modeling could attempt a **Log-Transformation** of the Anxiety Score to account for the floor effect (many people scoring 0)."
4. **Non-Linearity** : Regression looks for straight lines. If the relationship is a "curve" (e.g., moderate stress is good, but too little or too much is bad), OLS will report a 0 correlation.

